

PinPoint: Synchronization-Aware and Explainable Audio–Visual Deepfake Detection

Author information withheld for double-anonymized review

Abstract

Audio–visual deepfakes create a forensic challenge because a detector must identify manipulated media while also exposing evidence that can be inspected by human reviewers. Existing multimodal detectors often fuse acoustic and visual features without explicitly constraining whether the learned interaction reflects temporal synchrony between speech and facial motion. This paper presents PinPoint, a synchronization-aware audio–visual detector designed to make cross-modal alignment both predictive and auditable. PinPoint combines an explainability-compatible ResNet-18 visual encoder, a convolutional Gated Recurrent Unit audio encoder, stacked gated cross-attention blocks, a three-stage curriculum, and a multi-component synchronization loss that encourages aligned real samples to produce diagonal, concentrated, and temporally smooth audio-to-video attention maps. On a unified benchmark constructed from LAV-DF and FakeAVCeleb, PinPoint achieves 97.47% accuracy over 26,097 test samples, with macro F1 of 0.9680 and fake-class F1 of 0.9825. Ablation experiments show that removing the gate, synchronization loss, or curriculum reduces accuracy to 58.35%, 75.47%, and 81.44%, respectively. A multimodal explainable artificial intelligence analysis using Integrated Gradients, Layer-wise Relevance Propagation, SHAP, LIME, attention visualization, counterfactuals, and Testing with Concept Activation Vectors links the predictions to mouth-region, acoustic-feature, and lip-sync evidence. These results indicate that synchrony-aware supervision can improve deepfake detection accuracy while making model behavior easier to audit.

Keywords: Deepfake detection, Audio–visual fusion, Explainable artificial intelligence, Synchronization loss, Cross-attention, Curriculum learning

1. Introduction

Generative models can now alter speech, facial identity, and lip motion with a realism that makes manipulated videos difficult to assess by visual inspection alone. This changes deepfake detection from a simple binary classification problem into a forensic decision problem: the detector must be accurate, but it must also provide evidence that can be reviewed when the output affects journalism, legal evidence, political communication, or personal reputation. Recent surveys of passive multimodal deepfake detection emphasize the same pressure point: robustness, generalization, and interpretability remain open concerns when detectors are exposed to new manipulation pipelines and post-processing conditions [1].

Many early detectors exploited visual artifacts such as blending boundaries, texture mismatch, lighting inconsistency, or high-frequency synthesis traces. These cues can work well when training and test data share the same generation process, but they become less reliable as synthesis pipelines improve and compression removes fragile artifacts. Audio-only detectors face a similar problem because spectral synthesis artifacts may disappear as voice conversion and text-to-speech systems improve. Audio-visual deepfakes, however, also create a more structural cue: the relationship between what is spoken and how the face moves while speaking.

Synchronization cues are attractive because they are tied to the physics and timing of human communication. A manipulated video may preserve plausible frames and plausible audio independently while still disrupting the temporal correspondence between phonetic events, mouth openings, and facial motion. Prior work has shown that audio-visual dissonance, lip motion, and multimodal representation learning can improve detection [4–6, 8]. Yet fusion by itself does not guarantee synchrony-aware reasoning. A cross-attention model can still learn shortcut correlations if the training objective only rewards final real/fake classification.

This limitation matters for forensic use. If a detector marks a video as fake, an examiner should be able to ask whether the model relied on mouth-region evidence, acoustic timing, audio-visual mismatch, or unrelated visual artifacts. Post-hoc saliency maps can help, but they do not prove that the model learned a synchronization concept. PinPoint addresses this gap by making synchronization part of the optimization target and then auditing the trained model with multiple explanation methods.

We introduce PinPoint, an interpretable audio-visual architecture for

38 deepfake detection. Figure 1 summarizes the proposed framework. PinPoint
 39 processes 30-frame video clips and Mel-frequency cepstral coefficient (MFCC)
 40 audio sequences with modality-specific encoders, fuses them with gated cross-
 41 attention, and trains the attention structure with a synchronization loss ap-
 42 plied to aligned real samples. The same attention map used for prediction
 43 therefore receives a task-specific interpretation: diagonal structure indicates
 44 that audio timesteps attend to temporally corresponding video frames, while
 45 diffuse or off-diagonal structure signals weaker audio–visual correspondence.

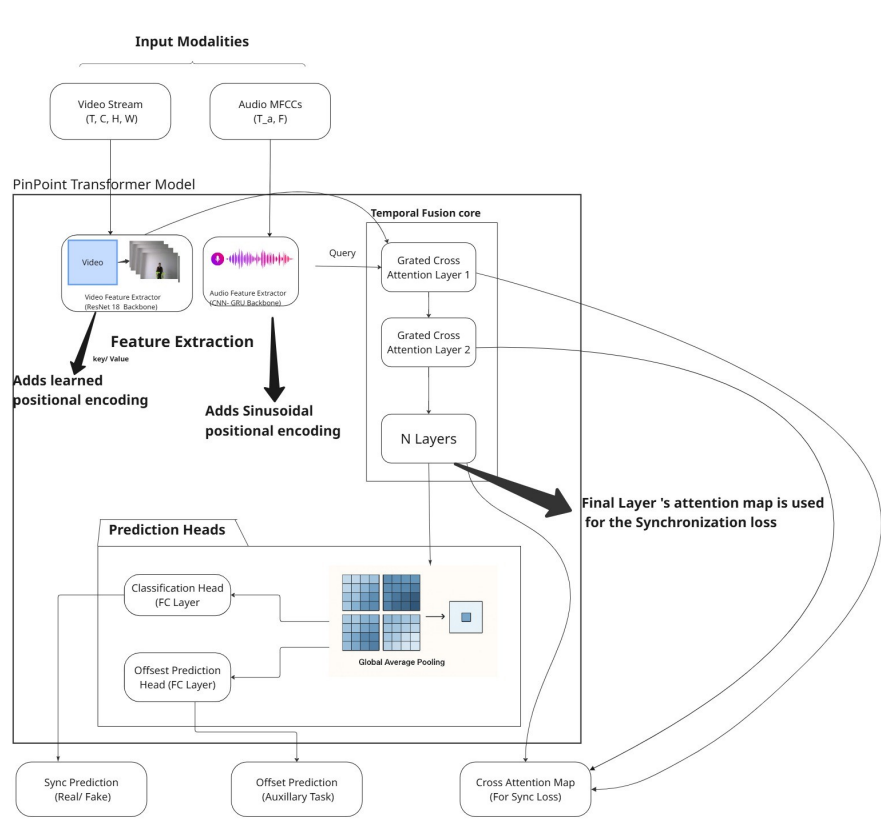


Figure 1: Overview of the PinPoint framework. The model receives aligned audio and video inputs, extracts modality-specific features, applies gated audio-to-video cross-attention, and trains with a curriculum that first encourages audio reliance, then synchronization learning, and finally full multimodal fusion. The synchronization loss shapes attention maps toward diagonal, smooth, and concentrated structures for aligned real samples.

46 The contributions of this paper are as follows:

- 47 1. We propose an audio–visual deepfake detector that uses gated cross-
48 attention to model temporal correspondence between speech features
49 and video frames rather than treating multimodal fusion as uncon-
50 strained feature concatenation.
- 51 2. We introduce a synchronization-aware training objective that combines
52 soft target matching, diagonal dominance, and temporal smoothness
53 to make cross-modal attention maps both discriminative and inter-
54 pretable.
- 55 3. We train the detector with a three-stage curriculum that first masks
56 video evidence, then emphasizes synchronization, and finally optimizes
57 full multimodal classification.
- 58 4. We evaluate model behavior with a multimodal explainable artificial
59 intelligence (XAI) suite that includes local attribution, attention anal-
60 ysis, counterfactual perturbation, and concept activation analysis.
- 61 5. We validate the design on a unified LAV-DF and FakeAVCeleb bench-
62 mark, where PinPoint achieves 97.47% accuracy and ablation experi-
63 ments show that removing the gate, synchronization loss, or curriculum
64 sharply reduces performance.

65 The remainder of the paper is organized as follows. Section 2 reviews
66 audio–visual deepfake detection and explainable multimedia forensics. Sec-
67 tion 3 presents the PinPoint architecture and training objective. Section 4
68 describes the explanation protocol. Section 5 reports the experimental setup,
69 classification results, ablations, and explanation analysis. Section 6 discusses
70 implications and limitations, and Section 7 concludes the paper.

71 2. Related work

72 2.1. Audio–visual deepfake benchmarks and synchronization cues

73 Multimodal deepfake detection depends on datasets that contain both
74 acoustic and visual manipulations. FakeAVCeleb introduced a multimodal
75 benchmark containing real videos, fake audio with real video, real audio with
76 fake video, and jointly manipulated audio–video samples [2]. LAV-DF later
77 focused on content-driven audio–visual forgery and temporal localization,
78 making it useful for studying cases where only part of a clip is manipulated
79 [3]. These datasets expose a limitation of unimodal detectors: a video can
80 appear visually plausible while its speech and facial motion are no longer
81 temporally consistent.

Early synchronization-aware approaches used audio–visual dissonance and lip-region dynamics to detect mismatches [4, 5]. More recent systems fuse audio and visual streams with learned multimodal representations. Multimodaltrace uses separate audio and visual representations and combines them for deepfake classification [6]. Temporal feature prediction methods exploit the expectation that adjacent audio–visual segments should be mutually predictable in authentic videos [7]. AVSFF models lip synchronization and reports strong performance across several multimodal datasets [8]. These studies support the value of synchronization cues, but many methods still treat alignment as an implicit consequence of architecture rather than as a directly supervised property of the learned attention map.

2.2. Cross-modal fusion and shortcut learning

Modern multimodal detectors increasingly rely on cross-modal attention, distillation, and hierarchical aggregation. CAD aligns multimodal representations and uses distillation to improve video deepfake detection [9]. DiMoDif models differences in machine perception across modalities for detection and localization [10]. Emerging benchmarks such as MAVOS-DD further show that detectors can lose performance under open-set language and generator shifts [11]. This creates a need for training objectives that discourage brittle shortcuts and push the model toward stable cross-modal evidence.

PinPoint follows this direction but differs in the role assigned to attention. Instead of using attention only as a flexible fusion operator, PinPoint makes the final audio-to-video attention map part of the supervised target. For aligned real samples, the desired attention structure is known approximately: each audio timestep should concentrate energy near the corresponding video frame and should change smoothly over time. The synchronization loss encodes this expectation directly.

2.3. Explainability for multimedia forensics

Explainability is a practical requirement for multimedia forensics because a correct label does not by itself justify a decision. Feature attribution methods such as Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), Integrated Gradients, Layer-wise Relevance Propagation (LRP), and Gradient-weighted Class Activation Mapping (Grad-CAM) provide different views of model evidence [13–17]. Concept-based methods such as Testing with Concept Activation Vectors (TCAV) test whether human-defined concepts influence internal representations [18].

Deepfake-specific XAI research has begun moving from qualitative heatmaps toward quantitative explanation evaluation. Baldassarre et al. proposed metrics for explanation smoothness, locality, and sparsity in video deepfake detectors [19]. Tsigos et al. evaluated explanation methods by measuring whether highlighted regions affect detector decisions under perturbation [20]. Recent work in Ain Shams Engineering Journal has also emphasized unified explainable multimodal deepfake frameworks [12]. PinPoint builds on this trend by evaluating not only visual saliency but also audio attribution, attention structure, counterfactual sensitivity, and concept-level response to lip-sync error.

3. Materials and methods

3.1. Problem formulation and inputs

Given a video segment V and its corresponding audio representation A , the task is to predict a binary label $y \in \{0, 1\}$, where 0 denotes real and 1 denotes fake. Each video sample is represented by 30 RGB frames resized to 128×128 and normalized with ImageNet mean and standard deviation. Each audio sample is represented as a time sequence of 13-dimensional MFCC features. The model outputs a fake logit, an auxiliary audio–video offset prediction, and the final audio-to-video attention map used for synchronization analysis.

3.2. Dataset construction and preprocessing

The experiments use a unified preprocessed benchmark built from LAV-DF and FakeAVCeleb. The dataset keeps the available train, validation, and test split metadata instead of re-splitting samples locally. Table 1 reports the resulting split sizes and class balance. The test split contains 26,097 samples, with 6,906 real and 19,191 fake samples.

Table 1: Unified benchmark split sizes and class balance.

Split	Total	Real	Fake	Real/Fake ratio
Train	78,703	21,254	57,449	0.37
Validation	31,501	8,271	23,230	0.36
Test	26,097	6,906	19,191	0.36

During training, video augmentations include JPEG compression, random noise, Gaussian blur, low-resolution resizing artifacts, color jitter, and random erasing. These augmentations are intended to reduce dependence on low-level generator artifacts. A modality dropout probability of 0.15 randomly drops either audio or video features for a sample during training, forcing the model to preserve useful evidence from both streams. For real training samples, artificial audio offsets are inserted with probability 0.5 over a maximum offset range of five video frames. These offset labels supervise the auxiliary offset prediction head.

3.3. *PinPoint architecture*

PinPoint contains a visual encoder, an audio encoder, stacked gated cross-attention blocks, and two prediction heads. The visual encoder is an ImageNet-pretrained ResNet-18 [21]. To support gradient-based explanation methods, all in-place rectified linear unit operations and residual additions are patched into non-in-place operations. The final convolutional features are adaptively pooled and projected to a 256-dimensional frame embedding. Early ResNet layers are frozen during training.

The audio encoder contains two one-dimensional convolution layers followed by layer normalization and a Gated Recurrent Unit (GRU). The convolution layers map 13 MFCC channels to 64 and then 128 channels using kernel size 3 and padding 1. The GRU maps the sequence to the same 256-dimensional embedding space as the video stream. Video embeddings use learned positional encodings, while audio embeddings use sinusoidal positional encodings.

Fusion uses three gated cross-attention blocks. In each block, audio embeddings act as queries and video embeddings act as keys and values. The resulting attention map therefore answers a temporally meaningful question: for each audio timestep, which video frames support the current representation? Figure 2 illustrates the block. After cross-attention, a sigmoid gate filters the attended audio representation. The gated representation then passes through audio self-attention and a feed-forward network with a hidden dimension of $4d$, where $d = 256$.

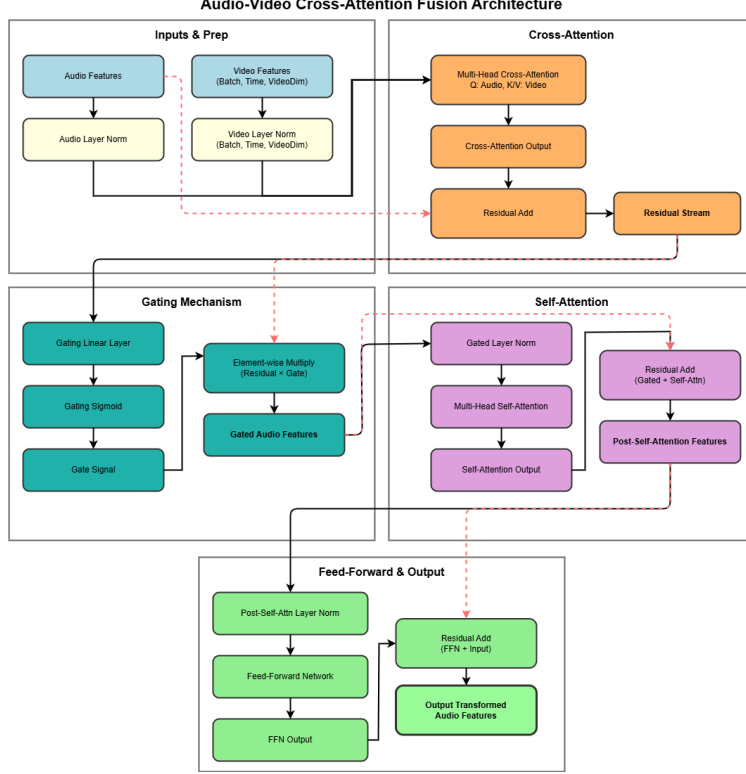


Figure 2: Gated cross-attention block. Audio features query video features, producing an audio-to-video attention map. A learned sigmoid gate filters the attended representation before self-attention and feed-forward refinement.

176 The final audio sequence is mean-pooled and passed to two heads. The
 177 classification head predicts the fake logit. The auxiliary offset head predicts
 178 one of $2M + 1$ offset classes, where $M = 5$ is the maximum frame offset.
 179 This auxiliary task gives the model an explicit training signal for temporal
 180 misalignment on real samples.

181 3.4. Synchronization-aware loss

182 The synchronization loss shapes the final cross-attention map for real sam-
 183 ples whose audio and video are aligned. Let $\mathbf{A} \in \mathbb{R}^{N \times M}$ denote the attention
 184 map between N audio timesteps and M video frames. The loss combines
 185 direct target matching, diagonal dominance, and temporal smoothness:

$$\mathcal{L}_{sync} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{diag} \mathcal{L}_{diag} + \lambda_{smooth} \mathcal{L}_{smooth}, \quad (1)$$

186 where $\lambda_{mse} = 1.0$, $\lambda_{diag} = 0.5$, and $\lambda_{smooth} = 0.2$. The first term matches
 187 the attention map to a soft diagonal target:

$$\mathcal{L}_{mse} = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M (A_{ij} - T_{ij})^2. \quad (2)$$

188 The target T_{ij} is a normalized Gaussian band around the expected cor-
 189 responding video frame:

$$T_{ij} = \frac{\exp(-(j - c_i)^2 / (2\sigma^2))}{\sum_{m=1}^M \exp(-(m - c_i)^2 / (2\sigma^2))}, \quad (3)$$

190 where $c_i = \lfloor i/r \rfloor$, $r = 2$ MFCC timesteps per video frame, and $\sigma = 2$.
 191 The second term penalizes attention energy outside a diagonal band of width
 192 $b = 2$:

$$\mathcal{L}_{diag} = 1 - \frac{\sum_{i=1}^N \sum_{|j-c_i| \leq b} A_{ij}}{\sum_{i=1}^N \sum_{j=1}^M A_{ij}}. \quad (4)$$

193 The third term discourages abrupt changes across neighboring audio timesteps
 194 and video frames:

$$\mathcal{L}_{smooth} = \frac{1}{N-1} \sum_{i=1}^{N-1} \sum_{j=1}^M (A_{ij} - A_{i+1,j})^2 + \frac{1}{M-1} \sum_{i=1}^N \sum_{j=1}^{M-1} (A_{ij} - A_{i,j+1})^2. \quad (5)$$

195 The loss is applied only to aligned real samples. Fake samples and arti-
 196 ficially offset real samples do not have a diagonal synchronization target, so
 197 they are excluded from this term.

198 3.5. Training objective and curriculum

199 The total loss combines binary classification, auxiliary offset prediction,
 200 and synchronization supervision:

$$\mathcal{L}_{total} = W_{class} \mathcal{L}_{class} + W_{offset} \mathcal{L}_{offset} + W_{sync} \mathcal{L}_{sync}, \quad (6)$$

201 where $W_{class} = 1.0$, $W_{offset} = 0.5$, and $W_{sync} = 3.0$. The classification
 202 term uses binary cross-entropy with logits and a positive-class weight com-
 203 puted from the real/fake composition of the training split. The offset term
 204 uses cross-entropy over 11 offset classes and is applied only to real samples.

205 Training follows a three-stage curriculum. In Phase 1, covering the first
 206 two epochs, 50–80% of video frames are randomly masked to force the model
 207 to learn useful audio evidence. In Phase 2, covering the next three epochs,
 208 synchronization learning receives emphasis and the offset loss is down-weighted
 209 by a factor of 0.1. In Phase 3, covering the remaining ten epochs, the model
 210 trains with full multimodal evidence and the complete composite objective.
 211 Table 2 summarizes the main training settings.

Table 2: Key implementation settings for PinPoint.

Parameter	Value
Video input	30 frames, 128×128 RGB
Audio input	13 MFCC coefficients per timestep
Visual encoder	ImageNet-pretrained ResNet-18
Audio encoder	Conv1D(64), Conv1D(128), LayerNorm, GRU(256)
Embedding dimension	256
Cross-attention heads / layers	8 / 3
Dropout	0.1
Batch size / epochs	4 / 15
Optimizer	AdamW
Learning rate / weight decay	1×10^{-4} / 1×10^{-4}
Learning-rate schedule	Cosine annealing
Gradient clipping	1.0 maximum norm
Modality dropout	0.15
Maximum offset	5 video frames
Offset probability	0.5 for real training samples

212 4. Explainable AI evaluation protocol

213 PinPoint is evaluated with complementary explanation tools because no
 214 single explanation method is sufficient for a multimodal forensic model. The
 215 XAI protocol tests four forms of evidence: local feature attribution, temporal
 216 attention structure, counterfactual sensitivity, and concept-level response.

217 4.1. Attribution and attention methods

218 Integrated Gradients and LRP produce gradient- and relevance-based
 219 attribution maps over video frames and MFCC features. SHAP estimates

220 feature contribution relative to a small background set, and LIME builds lo-
 221 cal surrogate explanations for both modalities. Attention analysis visualizes
 222 the final cross-attention map and attention rollout across stacked attention
 223 layers. These methods provide complementary checks: agreement between
 224 independent attribution methods increases confidence that the highlighted
 225 evidence is not an artifact of a single explainer.

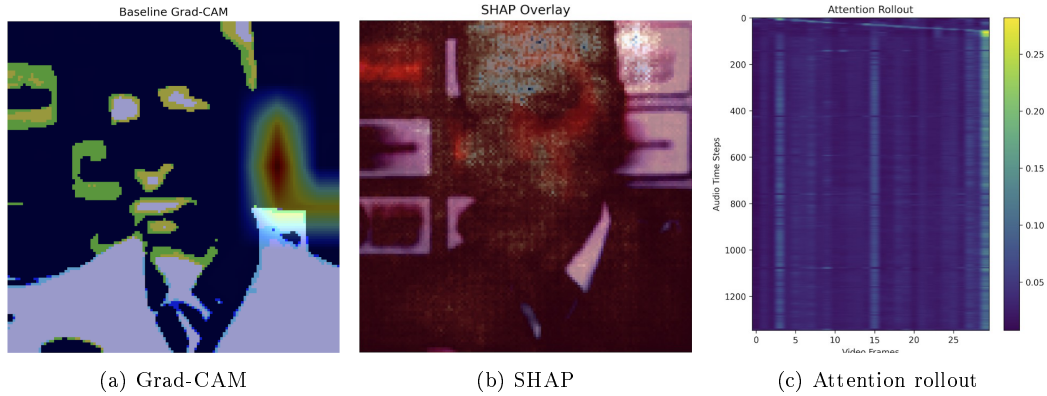


Figure 3: Example multimodal explanations for a fake sample. Grad-CAM and SHAP highlight visual evidence around the face and mouth region, while attention rollout exposes the temporal structure used during audio–visual fusion.

226 4.2. Counterfactual and concept analysis

227 Counterfactual analysis searches for small audio and video perturbations
 228 that change the predicted class. This provides a causal stress test: if changing
 229 the highlighted visual or acoustic evidence flips the decision, the explanation
 230 is more informative than a static heatmap alone. TCAV evaluates whether
 231 the model is sensitive to a human-defined lip-sync error concept by training
 232 concept activation vectors at selected internal layers and measuring whether
 233 movement in that concept direction changes the fake-class output.

234 4.3. Quantitative explanation metrics

235 The quantitative analysis reports sparsity and inter-technique consis-
 236 tency. Sparsity is measured with the Gini index:

$$G = \frac{\sum_{i=1}^n (2i - n - 1)x_i}{n \sum_{i=1}^n x_i}, \quad (7)$$

237 where x_i are sorted attribution values. Higher values indicate more con-
 238 centrated attribution. Inter-technique consistency is measured with Spear-
 239 man rank correlation between flattened attribution maps from different ex-
 240 planation methods:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}, \quad (8)$$

241 where d_i is the rank difference for feature i . Consistency is important
 242 because high sparsity alone is not necessarily desirable; a sparse explanation
 243 can still focus on unstable or irrelevant evidence.

244 5. Experiments and results

245 5.1. Main classification performance

246 Table 3 reports the full PinPoint model on the unified test split. The
 247 model correctly classifies 25,436 of 26,097 samples, giving 97.47% accuracy.
 248 Fake-class F1 reaches 0.9825, and real-class recall reaches 0.98. The high real-
 249 class recall is important because a forensic detector should avoid incorrectly
 250 accusing authentic content whenever possible.

Table 3: PinPoint performance on the unified LAV-DF and FakeAVCeleb test split.

Class / aggregate	Precision	Recall	F1-score	Support
Real	0.93	0.98	0.9535	6,906
Fake	0.99	0.97	0.9825	19,191
Macro average	0.96	0.98	0.9680	26,097
Weighted average	0.98	0.97	0.9700	26,097
Accuracy	–	–	97.47%	26,097

251 The confusion matrix in Table 4 shows that most errors are fake samples
 252 predicted as real. This pattern is preferable to an uncontrolled false-positive
 253 tendency, but it also indicates that future work should test more difficult ma-
 254 nipulation families and calibrate decision thresholds for the intended forensic
 255 setting.

Table 4: Confusion matrix for the full PinPoint model on the unified test split.

Ground truth	Predicted real	Predicted fake
Real	6,767	139
Fake	522	18,669

Table 5 places the result beside representative audio–visual detectors. This table should not be read as a strict leaderboard because the compared papers use different datasets, splits, and evaluation protocols. Its purpose is to contextualize the performance range of recent multimodal methods while keeping PinPoint’s primary evidence in the unified benchmark and ablation study.

Table 5: Contextual comparison with representative audio–visual deepfake detectors. Metrics are not directly comparable across rows because datasets and protocols differ.

Method	Dataset(s)	Accuracy	Precision	Recall	F1-score
PinPoint	Unified benchmark	97.47%	0.98	0.97	0.97
AVSFF [8]	FakeAVCeleb	99.73%	0.9971	0.9944	0.9907
AVSFF [8]	LAV-DF	97.86%	–	–	–
Multimodaltrace [6]	FakeAVCeleb	92.9%	–	–	0.945
TFP [7]	FakeAVCeleb	84.33%	–	–	–

5.2. Ablation study

The ablation study isolates the gate, synchronization loss, and curriculum while keeping the unified evaluation protocol fixed. Table 6 shows that each component contributes to performance. Removing the gate causes the largest degradation, reducing accuracy from 97.47% to 58.35%. Removing the synchronization loss reduces accuracy to 75.47%, and removing the curriculum reduces accuracy to 81.44%.

Table 6: Ablation study on the unified test split. Each variant removes one component from the full model.

Variant	Epochs	Accuracy	Fake F1	Fake precision	Fake recall
Full PinPoint	15	97.47%	0.9825	0.99	0.97
No synchronization loss	15	75.47%	0.8067	0.95	0.70
No gate	15	58.35%	0.6115	0.97	0.45
No curriculum	15	81.44%	0.8639	0.94	0.80

269 The gate is therefore not a minor architectural embellishment. It controls
270 how much cross-modal evidence enters the refined audio representation, and
271 removing it leaves the model with standard cross-attention that performs
272 poorly on fake recall. The synchronization loss is also essential: without
273 direct alignment supervision, the model retains high fake precision but misses
274 many manipulated samples. The curriculum result shows that training order
275 changes the learned decision behavior, not merely the speed of convergence.



Figure 4: LIME visualizations across model variants. The full model concentrates evidence more coherently around the mouth and face region, while ablated variants produce less stable or more diffuse visual explanations.

276 5.3. Quantitative explanation results

277 Table 7 reports explanation metrics for the full model and ablations.
 278 PinPoint produces moderate video sparsity, strong audio sparsity, and the
 279 highest inter-technique consistency among the evaluated variants. The con-

280 sistency result is the most important part of the table: multiple explanation
 281 methods agree more strongly for the full model, suggesting that the model’s
 282 evidence is more stable under different attribution lenses.

Table 7: Quantitative XAI metrics across model variants. Higher consistency indicates stronger agreement between explanation methods.

Model	Video Gini	Audio Gini	Consistency (Spearman)
Full PinPoint	0.631	0.928	0.514
No curriculum	0.803	0.919	0.450
No gate	0.753	0.760	0.372
No synchronization loss	0.708	0.883	0.436

283 Higher Gini values in the ablated models should not be interpreted as
 284 automatically better. A sparse explanation can be unhelpful if it focuses on
 285 unstable evidence. The full model’s combination of concentrated audio at-
 286 tribution and higher inter-technique consistency better matches the forensic
 287 requirement: explanations should be both localized and repeatable.

288 5.4. Case studies

289 Figure 5 compares attention maps for real and fake samples. The real
 290 sample shows a stronger diagonal pattern, indicating that audio timesteps
 291 attend to temporally corresponding video frames. The fake sample produces
 292 a more diffuse map, consistent with weaker audio–visual synchrony. This is
 293 the intended behavior of the synchronization loss: the attention map becomes
 294 an interpretable object rather than only a hidden fusion mechanism.

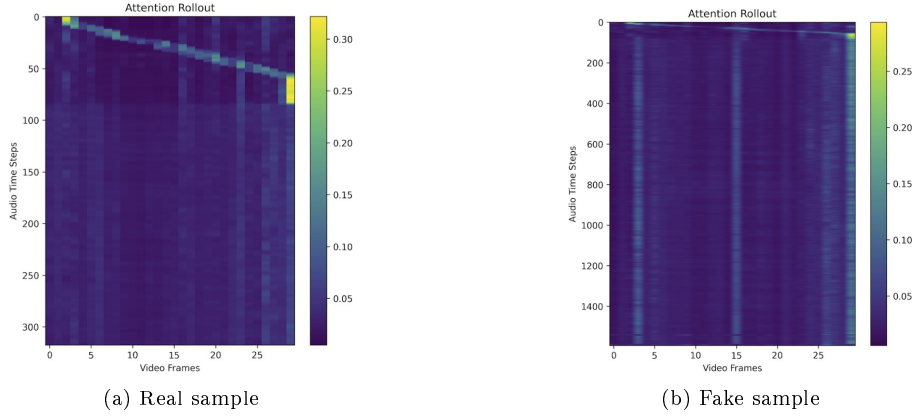


Figure 5: Cross-attention maps for representative real and fake samples. Real samples exhibit stronger diagonal alignment, while fake samples show more diffuse attention, indicating weaker audio–visual correspondence.

Figure 6 shows visual attribution on a fake sample. Grad-CAM emphasizes the mouth and lower-face region, while SHAP provides a complementary attribution view. These attributions agree with the synchronization framing because lip motion and mouth-region evidence are expected to carry much of the audio–visual mismatch signal.

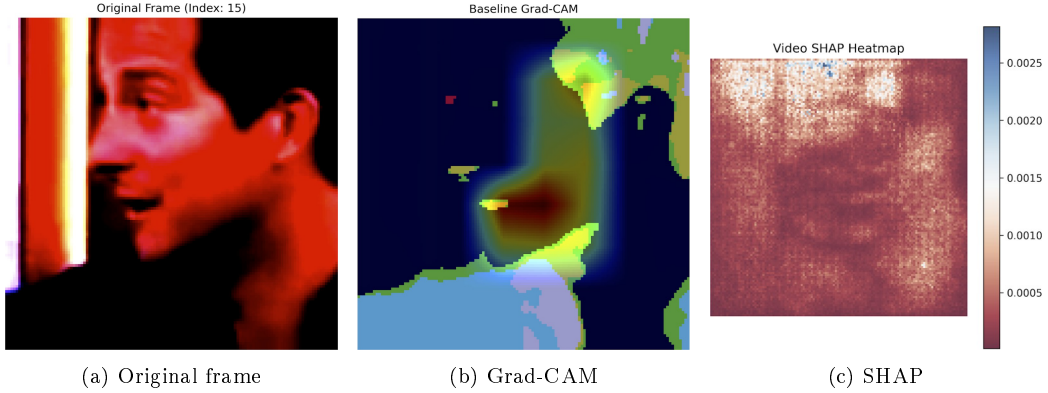


Figure 6: Attribution case study. Visual explanation methods identify face and mouth-region evidence that supports the fake prediction, complementing the temporal evidence from cross-attention.

Counterfactual analysis tests whether changing the highlighted evidence can alter the prediction. Figure 7 shows a perturbation that changes a fake

302 prediction toward real. The difference map localizes the visual evidence re-
 303 sponsible for the decision shift, providing a stronger check than a purely
 304 descriptive saliency map.

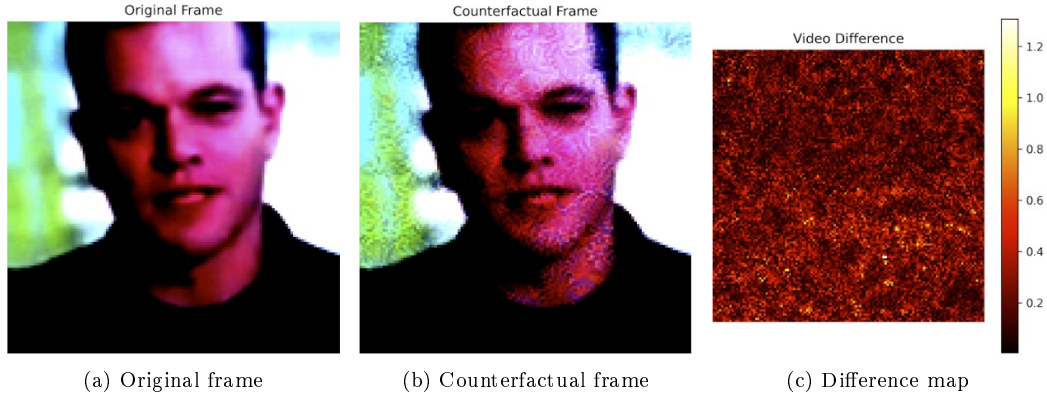


Figure 7: Counterfactual explanation. A small perturbation changes the model decision, and the difference map localizes the visual evidence most responsible for the decision shift.

305 The TCAV analysis further tests concept-level behavior. Figure 8 shows
 306 high sensitivity to the lip-sync error concept at the gated attention layer
 307 and classification head. These results support the claim that synchronization
 308 information influences both the internal representation and the final decision.

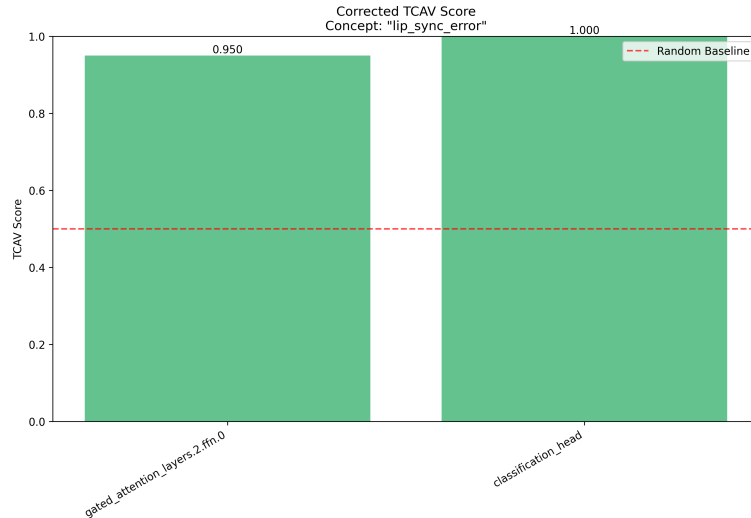


Figure 8: TCAV scores for the lip-sync error concept. Scores above the random baseline indicate that the concept influences internal representations and the fake-class decision.

309 6. Discussion

310 The results support the central design claim of PinPoint: synchrony-
311 aware supervision improves both classification performance and explanation
312 quality. The ablation study shows that the gate, synchronization loss, and
313 curriculum are all necessary. The no-gate model performs especially poorly
314 on fake recall, which indicates that standard cross-attention is not enough
315 for this task. The no-synchronization-loss model performs better than no-
316 gate but still loses more than 22 percentage points of accuracy, showing that
317 explicitly shaping attention structure contributes substantially to the final
318 detector.

319 The XAI results also show why synchronization should be treated as
320 part of the model design rather than as a post-hoc interpretation. In the full
321 model, the attention map has a defined forensic meaning because it is trained
322 to match expected audio-visual timing in aligned real samples. Attribution,
323 counterfactual, and TCAV analyses then serve as checks on whether this
324 learned structure agrees with visual, acoustic, and concept-level evidence.
325 This layered explanation strategy is more informative than presenting a single
326 heatmap after training.

327 PinPoint is most useful in settings where a detector must support review
328 rather than simply return a label. A forensic analyst can inspect the pre-
329 dicted class, the attention structure, the visual and audio attributions, and
330 the counterfactual sensitivity. These outputs do not eliminate the need for
331 human judgment, but they make the model’s decision pathway more trans-
332 parent.

333 Several limitations remain. First, the current evaluation uses a unified
334 benchmark built from curated datasets. A stronger generalization protocol
335 should train on one dataset and test on another, and should include in-
336 the-wild compression and social-media transformations. Second, the model
337 uses fixed 30-frame segments, which may miss longer-range inconsistencies.
338 Third, the full XAI suite is computationally expensive because SHAP, coun-
339 terfactual generation, and gradient-based attribution require additional for-
340 ward or backward passes. Fourth, the available logs do not include repeated
341 runs or confidence intervals, so this paper avoids claims of statistical signif-
342 icance. Future work should add multi-seed evaluation, calibration analysis,
343 and cross-dataset testing.

344 7. Conclusions

345 This paper presented PinPoint, a synchronization-aware and explainable
346 audio-visual deepfake detector. The model combines an XAI-compatible
347 ResNet-18 visual encoder, a CNN-GRU audio encoder, gated cross-attention,
348 curriculum training, and a synchronization loss that shapes attention maps
349 toward diagonal, smooth, and concentrated structures for aligned real sam-
350 ples.

351 On the unified LAV-DF and FakeAVCeleb benchmark, PinPoint achieved
352 97.47% accuracy and 0.9680 macro F1 over 26,097 test samples. Ablation ex-
353 periments showed that removing the gate, synchronization loss, or curriculum
354 sharply degrades performance, confirming that synchronization-aware fusion
355 is central to the detector. The XAI analysis showed that the full model pro-
356 duces more consistent explanations and that lip-sync error concepts influence
357 internal representations and the final prediction.

358 The broader conclusion is that interpretability in deepfake detection should
359 not be treated only as a post-hoc add-on. By making audio-visual synchrony
360 part of the training objective and then auditing the resulting model with at-
361 tribution, attention, counterfactual, and concept-based tools, PinPoint pro-
362 vides a more transparent path toward forensic multimedia analysis.

363 Declarations

364 **Funding.** This research did not receive any specific grant from funding
365 agencies in the public, commercial, or not-for-profit sectors.

366 **Competing interests.** The authors declare that they have no known
367 competing financial interests or personal relationships that could have ap-
368 peared to influence the work reported in this paper.

369 **Data availability.** LAV-DF and FakeAVCeleb are publicly described
370 datasets. Derived preprocessed tensors and trained model artifacts can be
371 made available by the authors upon reasonable request, subject to the licenses
372 and access terms of the original datasets.

373 **Code availability.** The implementation is available in the project repos-
374 itory during review and can be released with the final manuscript subject to
375 institutional approval.

376 **CRedit author statement.** Author contribution details are provided
377 in the separate title page file to preserve double-anonymized review.

378 **Declaration of generative AI and AI-assisted technologies.** A
379 disclosure statement is provided in the separate title page file for the authors
380 to review and adjust before submission.

381 **References**

- 382 [1] Nguyen-Le, H.H., Tran, V.T., Nguyen, D.T., Le-Khac, N.A. Passive
383 deepfake detection across multi-modalities: A comprehensive survey.
384 arXiv preprint arXiv:2411.17911, 2024.
- 385 [2] Khalid, H., Tariq, S., Kim, M., Woo, S.S. FakeAVCeleb: A novel audio-
386 video multimodal deepfake dataset. arXiv preprint arXiv:2108.05080,
387 2021.
- 388 [3] Cai, Z., Stefanov, K., Dhall, A., Hayat, M. Do you really mean that?
389 Content driven audio-visual deepfake dataset and multimodal method
390 for temporal forgery localization. In: 2022 International Conference on
391 Digital Image Computing: Techniques and Applications (DICTA), pp.
392 1–10, 2022. doi:10.1109/DICTA56598.2022.10034605.
- 393 [4] Chugh, K., Gupta, P., Dhall, A., Subramanian, R. Not made for each
394 other: Audio-visual dissonance-based deepfake detection and localiza-
395 tion. In: Proceedings of the 28th ACM International Conference on
396 Multimedia, pp. 439–447, 2020. doi:10.1145/3394171.3413700.
- 397 [5] Haliassos, A., Vougioukas, K., Petridis, S., Pantic, M. Lips don’t lie:
398 A generalisable and robust approach to face forgery detection. In: Pro-
399 ceedings of the IEEE/CVF Conference on Computer Vision and Pattern
400 Recognition, pp. 5039–5049, 2021. doi:10.1109/CVPR46437.2021.00500.
- 401 [6] Raza, M.A., Malik, K.M. Multimodaltrace: Deepfake detection using
402 audiovisual representation learning. In: Proceedings of the IEEE/CVF
403 Conference on Computer Vision and Pattern Recognition Workshops,
404 pp. 993–1000, 2023. doi:10.1109/CVPRW59228.2023.00048.
- 405 [7] Gao, Y., Wang, X., Zhang, Y., Zeng, P., Ma, Y. Temporal feature predic-
406 tion in audio-visual deepfake detection. *Electronics* 13(17), 3433, 2024.
407 doi:10.3390/electronics13173433.

- 408 [8] Javed, M., Zhang, Z., Dahri, F.H., Laghari, A.A., Krajcik, M., Al-
409 madhor, A. Audio-visual synchronization and lip movement analysis for
410 real-time deepfake detection. *International Journal of Computational*
411 *Intelligence Systems* 18, 170, 2025. doi:10.1007/s44196-025-00911-7.
- 412 [9] Du, Y., Wang, Z., Luo, Y., Piao, C., Yan, Z., Li, H., Yuan, L.
413 CAD: A general multimodal framework for video deepfake detection via
414 cross-modal alignment and distillation. *arXiv preprint arXiv:2505.15233*,
415 2025.
- 416 [10] Koutlis, C., Papadopoulos, S. DiMoDif: Discourse modality-information
417 differentiation for audio-visual deepfake detection and localization.
418 *arXiv preprint arXiv:2411.10193*, 2024.
- 419 [11] Croitoru, F.A., Hondru, V., Popescu, M., Ionescu, R.T., Khan, F.S.,
420 Shah, M. MAVOS-DD: Multilingual audio-video open-set deepfake de-
421 tection benchmark. *arXiv preprint arXiv:2505.11109*, 2025.
- 422 [12] Kaya, M., Alhajj, R. An integrated explainable framework
423 for multimodal deepfake detection across image, audio, and
424 video data. *Ain Shams Engineering Journal* 17(7), 104233, 2026.
425 doi:10.1016/j.asej.2026.104233.
- 426 [13] Ribeiro, M.T., Singh, S., Guestrin, C. Why should I trust you? Explain-
427 ing the predictions of any classifier. In: *Proceedings of the 22nd ACM*
428 *SIGKDD International Conference on Knowledge Discovery and Data*
429 *Mining*, pp. 1135–1144, 2016. doi:10.1145/2939672.2939778.
- 430 [14] Lundberg, S.M., Lee, S.I. A unified approach to interpreting model pre-
431 dictions. In: *Advances in Neural Information Processing Systems* 30, pp.
432 4765–4774, 2017.
- 433 [15] Sundararajan, M., Taly, A., Yan, Q. Axiomatic attribution for deep net-
434 works. In: *Proceedings of the 34th International Conference on Machine*
435 *Learning*, pp. 3319–3328, 2017.
- 436 [16] Bach, S., Binder, A., Montavon, G., Klauschen, F., Muller, K.R.,
437 Samek, W. On pixel-wise explanations for non-linear classifier decisions
438 by layer-wise relevance propagation. *PLOS ONE* 10(7), e0130140, 2015.
439 doi:10.1371/journal.pone.0130140.

- 440 [17] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra,
441 D. Grad-CAM: Visual explanations from deep networks via gradient-
442 based localization. In: Proceedings of the IEEE International Conference
443 on Computer Vision, pp. 618–626, 2017. doi:10.1109/ICCV.2017.74.
- 444 [18] Kim, B., Wattenberg, M., Gilmer, J., Cai, C., Wexler, J., Viegas,
445 F., Sayres, R. Interpretability beyond feature attribution: Quantita-
446 tive testing with concept activation vectors (TCAV). In: Proceedings of
447 the 35th International Conference on Machine Learning, pp. 2668–2677,
448 2018.
- 449 [19] Baldassarre, F., Debard, Q., Pontiveros, G.F., Wijaya, T.K. Quan-
450 titative metrics for evaluating explanations of video DeepFake detec-
451 tors. In: Proceedings of the British Machine Vision Conference, 2022.
452 arXiv:2210.03683.
- 453 [20] Tsigos, K., Apostolidis, E., Baxevas, S., Papadopoulos, S., Mezaris,
454 V. Towards quantitative evaluation of explainable AI methods for
455 deepfake detection. In: Proceedings of the 3rd ACM International
456 Workshop on Multimedia AI against Disinformation, pp. 37–45, 2024.
457 doi:10.1145/3643491.3660292.
- 458 [21] He, K., Zhang, X., Ren, S., Sun, J. Deep residual learning
459 for image recognition. In: Proceedings of the IEEE Conference
460 on Computer Vision and Pattern Recognition, pp. 770–778, 2016.
461 doi:10.1109/CVPR.2016.90.